

Quantum Pyramid

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Difficulty Level : Hard

Quantum Pyramid

Raouf is building his own quantum computer, but as you may know, qubits are really hard to implement and very unstable. For his first prototype — the **SpeedMachine** — he uses qubits alongside classical bits. Each cell of the pyramid-shaped machine is a *register* consisting of 31 classical bits and 1 qubit at the least-significant position. The 31 classical bits and the qubit together represent a 32-bit integer, where the qubit holds the LSB. A qubit behaves like a normal bit, except it can also be in a **superposition** of 0 and 1 simultaneously (denoted $=$).

You can see a SpeedMachine with 6 input integers in the figure below:

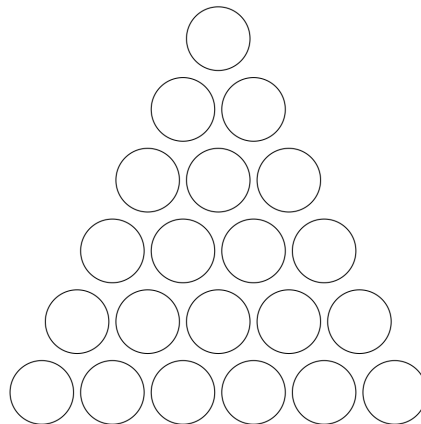


Figure 1: Quantum Pyramid Structure

The SpeedMachine of size n has n rows. The bottom row (row 1) is loaded with the input sequence of n integers. Each register in every other row reads the two registers directly below it and computes their **maximum over the 31 classical bits only** (the qubit is not part of the comparison). The qubit of the resulting register is then set as follows:

- If the two registers below have **different** classical bit values, the qubit simply follows the qubit of whichever register had the larger classical value.
- If the two registers below have **equal** classical bit values, then:
 - if both qubits below are 0, the new qubit is 0;
 - if both qubits below are 1, the new qubit is 1;
 - otherwise (at least one qubit is $=$ or the two qubits differ), the new qubit is $=$ (superposition).

When a register is **observed** (read), its value collapses as follows: if the qubit is 0 or 1, the observed integer is exactly $(c \ll 1) \mid q$, where c is the 31-bit classical value and $q \in \{0, 1\}$. If the qubit is $=$, the register can collapse to either $(c \ll 1)$ or $(c \ll 1) \mid 1$, and **both** possible values are reported.

Input

The first line contains two integers n and Q — the size of the SpeedMachine and the number of queries.

The second line contains n integers $a_1, a_2, \dots, a_n$ — the input sequence loaded into row 1.

Each of the next Q lines contains two integers k and r — a query asking for the value stored in the register at row r , column k .

Output

For each query, print either one integer (if the qubit is not in superposition) or two integers separated by a space in increasing order (if the qubit is in superposition $=$), representing all possible observed values.

Constraints

- $1 \leq n \leq 10^5, \quad 1 \leq Q \leq 10^5$
- $0 \leq a_i < 2^{32}$
- $1 \leq r \leq n, \quad 1 \leq k \leq n - r + 1$

Example

Input:

```
5 3
2 3 1 2 1
1 4
2 2
4 2
```

Output:

```
2 3
3
2
```

Explanation

Each input value is split into 31 classical bits (upper part) and 1 qubit (LSB):

- $a_1 = 2 = 10_2$: classical = 1, qubit = 0
- $a_2 = 3 = 11_2$: classical = 1, qubit = 1
- $a_3 = 1 = 01_2$: classical = 0, qubit = 1
- $a_4 = 2 = 10_2$: classical = 1, qubit = 0
- $a_5 = 1 = 01_2$: classical = 0, qubit = 1

Building the pyramid upward (notation: $c[q]$ where c = classical bits, q = qubit):

Row 2: - $k = 1$: $\max(1, 1) = 1$, equal, qubits 0 and 1 differ $\Rightarrow 1[=]$ - $k = 2$: $\max(1, 0) = 1$, differ, inherit qubit of $a_2 \Rightarrow 1[1]$ - $k = 3$: $\max(0, 1) = 1$, differ, inherit qubit of $a_4 \Rightarrow 1[0]$ - $k = 4$: $\max(1, 0) = 1$, differ, inherit qubit of $a_4 \Rightarrow 1[0]$

Row 3: - $k = 1$: $\max(1, 1) = 1$, equal, qubits = and 1 $\Rightarrow 1[=]$ - $k = 2$: $\max(1, 1) = 1$, equal, qubits 1 and 0 differ $\Rightarrow 1[=]$ - $k = 3$: $\max(1, 1) = 1$, equal, qubits 0 and 0 $\Rightarrow 1[0]$

Row 4: - $k = 1$: $\max(1, 1) = 1$, equal, qubits = and = $\Rightarrow 1[=]$ - $k = 2$: $\max(1, 1) = 1$, equal, qubits = and 0 $\Rightarrow 1[=]$

Now answering the queries:

- **Query 1** ($k = 1, r = 4$): register $1[=]$, classical = 1, qubit == $\Rightarrow$ observed values 2 and 3. Output: **2 3**
- **Query 2** ($k = 2, r = 2$): register $1[1]$, classical = 1, qubit = 1 $\Rightarrow$ observed value 3. Output: **3**
- **Query 3** ($k = 4, r = 2$): register $1[0]$, classical = 1, qubit = 0 $\Rightarrow$ observed value 2. Output: **2**